Problems:

1. Find the mean, mode, median and range for the given dataset. {2, 4, 6, 8, 2, 10, 12}

Sol: Mean $= \frac{2+4+6+8+2+10+12}{7} = \frac{44}{7} = 6.2$

Median (odd no. of dataset) $= \left[\frac{n+1}{2}\right]^{th}$ term $= \left(\frac{7+1}{2}\right)^{th} = \left(\frac{8}{2}\right)^{th}$ term

$= (4^{th})$ term

$= 8$.

Mode:

number	frequency
2	2
4	1
6	1
8	1
10	1
12	1

Mode = 2.

Range = maximum value − minimum value

$= 12 - 2 = 10$.

2. Find mean, mode, median, range for the dataset {5, 3, 1, 6, 8, 9}.

Sol: Mean $= \frac{5+3+1+6+8+9}{6} = \frac{32}{6} = \frac{16}{3} = 5.3$

Median (even no. of dataset) $= \frac{\left[\frac{n}{2}\right]^{th} \text{term} + \left[\frac{n}{2}+1\right]^{th} \text{term}}{2}$

$= \frac{\left(\frac{6}{2}\right)^{th} \text{term} + \left(\frac{6}{2}+1\right)^{th} \text{term}}{2} = \frac{(3)^{th} \text{term} + (4^{th}) \text{term}}{2}$

$= \frac{1+6}{2} = \frac{7}{2} = 3.5$

Mode = 5, 3, 1, 6, 8, 9.

Range = 9 − 1 = 8.

3. Find Mean, Mode, Median, Range from the dataset given: {3, 9, 12, 2, 3, 7, 16, 5}.

sol: Mean = $\frac{3+9+12+2+3+7+16+5}{8} = \frac{57}{8} = 7.125$

Median = (even no. of dataset) $\frac{\left[\frac{n}{2}\right]^{th} \text{ term} + \left[\frac{n}{2}+1\right]^{th} \text{ term}}{2} = \frac{(4^{th}) \text{term} + (5^{th}) \text{term}}{2}$

$= \frac{2+3}{2} = \frac{5}{2} = 2.5$

Mode:

number	frequency
3	2
9	1
12	1
2	1
7	1
16	1
5	1

Mode = 3

Range = 16 − 2 = 14.

Types of Big data:

i) Structured.

ii) Unstructured.

iii) Semi-Structured.

i) Structured: Highly organized, quantitative data (eg: SQL databases, spreadsheets, csv files) that fits neatly into rows and columns.

ii) Unstructured: Raw, unformatted data (eg: text, emails, social media post, audio, video) that requires advanced parsing and NLP to analyze.

iii) Semi Structured: Data that doesn't reside in a relational database but has some organizational properties. eg: JSON, XML, NOSQL databases).

Properties of Big data (The 5 V's):

1. Volume.
2. Velocity.
3. Variety.
4. Veracity.
5. Value.

Volume: Large volume of data generated from many sources daily.
Eg: business process, social media.
facebook generates 4.5 billion times "like" button.

Velocity: The speed at which new data is generated and must be processed.
eg: real time streaming feeds.

Variety: Big data can be structured, unstructured, semi structured. It collects from different sources.

Veracity: It is the process of being able to handle and manage data efficiently. The quality, reliability and trustworthiness of data.
Eg: facebook posts with hashtags.

Value: It is valuable and reliable data that we store, process and analyze.

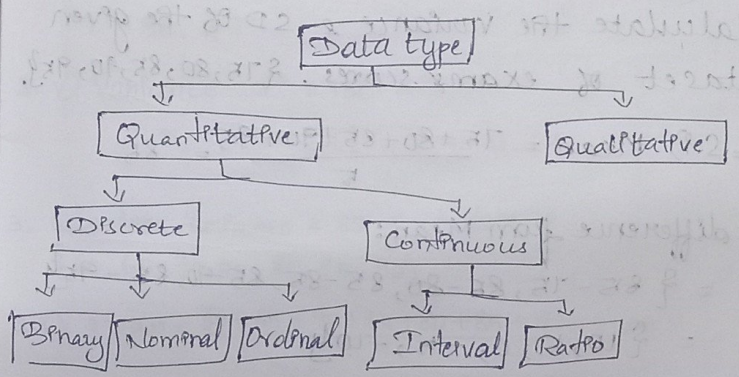
Big Data	Data Science
* Handling large data.	* Analyzing data.
* Hadoop, Spark, Flink are the tools.	* SAS, R, python are the tools.

Big Data	Datascience.
* Industry: e-commerce, security services, telecommunication.	* Industry: sales, Image recognition, advertisement, risk analytics.
* Responsibility: Proces huge volumes of data and genelate insights.	* Reponsibility: Understand pattern within data and make decisions.

Data Objects: * A data objects represents an entity.
* Data objects are described by attributes.
* Database rows → data objects; columns → attributes.

Attribute: a datafield, representing a characteristic or feature of data object.

Types: * Nominal.
* Binary.
* Ordinal.
* Continuous & discontinuous.
* Numeric: quantitative
* Interval-scaled.
* Ratio scaled.



Data Visualization: It is the graphical representation of information and data. By using visual elements like charts, graphs and maps, data visualization tools provide an accessible way to see & understand trends, outliers, and patterns in data.

Basic Statistics description of data:

1. Mean.
2. Median.
3. Mode.
4. Range.
5. Standard deviation (σ).
6. Variance (σ^2).

Steps to find Variance & standard deviation:

step 1: Calculate the mean value.
step 2: Calculate the difference from Mean.
step 3: Find the square of difference.
step 4: Calculate the mean of square value.
step 5: Calculate standard deviation.

$$SD = \sqrt{\text{variance}}$$

Problems:

1. Calculate the variance & SD of the given dataset of exam scores. {75, 80, 85, 90, 95}

Sol: S.I) Mean $= \frac{75+80+85+90+95}{5} = 85$

S.II) difference from Mean:

$= \{85-75, 85-80, 85-85, 85-90, 85-95\}$

$= \{10, 5, 0, -5, -10\}$

S.III) Square of difference:

$= \{100, 25, 0, 25, 100\}$.

S.iv) Mean of square value: $= \frac{100+25+0+25+100}{5} = \frac{250}{5}$

$= 50$

S. v) Variance $\sigma^2 = 50$.

Standard deviation $= \sigma = \sqrt{50} = 7.07$.

2. Calculate variance & SD for a given set.

{16, 25, 32, 34, 45, 51}.

S.i) Mean $= \frac{16+25+32+34+45+51}{6} = \frac{203}{6} = 33.8$

S.ii) difference from mean:

$= \{16-33.8, 25-33.8, 32-33.8, 34-33.8, 45-33.8, 51-33.8\}$

$= \{-17.8, -8.8, -1.8, 0.2, 11.2, 17.2\}$

S.iii) Square of difference:

$= \{316.84, 77.44, 3.24, 0.04, 125.44, 295.84\}$

S. iv) Mean of square value $= \frac{316.84+77.44+3.24+0.04+125.44+295.84}{6}$

$= 136.47$.

S. v) Variance $\sigma^2 = 136.47$

S.D $= \sigma = \sqrt{136.47} = 11.682$.

3. Calculate Variance & SD given dataset:

{60, 65, 70, 75, 80, 85, 90}.

S.i) Mean $= \frac{60+65+70+75+80+85+90}{7} = 75$.

S.ii) difference from mean:

{-15, -10, -5, 0, 5, 10, 15}

S.PPP) {225, 100, 25, 0, 25, 100, 225}

S.M) 100.

S.V) Variance $\sigma^2 = 100$, S.D = $\sigma = \sqrt{100} = 10$.

Data Visualization:

1. Bar chart:

```
import matplotlib.pyplot as plt
#Dataset
prog_lang = ['c', 'c++', 'Java', 'Python', 'PHP', 'Other']
emp_freq = [23, 18, 35, 30, 32, 15].
#Bar graph creation:
fig, ax = plt.subplots(figsize=(10,5))
plt.bar(prog_lang, emp_freq)
plt.title("Competent in a programming language")
plt.xlabel("Programming language")
plt.ylabel("Frequency")
plt.show().
```

Measuring data similarity & dissimilarity:

Similarity:

* The similarity measure is usually expressed as a binary value.
* It is often expressed as a no. between zero and one by conversion.
* zero means low similarity (the data objects are dissimilar).
* One means high similarity (the data objects are very similar).

Dissim

* Using
diffe
cluste

Distan

In da
between
Comm
1. Eucl
2. Man
3. Cosi
4. Ham

Euclide
points

Manhatt
distanc
points
d = |(

Cosine sim
two non
regardless of
their
cosine.

Hamming
equal len
symbols
* It is m
error det

Dissimilarity:

* Using knn algorithm. It describes how data objects are differ from one another. knn algorithm mostly used in clustering.

Distance measure: Distance measures are essential in data science for quantifying similarity or dissimilarity between data points, impacting algorithms in clustering.

Common distance measures:

1. Euclidean distance.
2. Manhattan distance.
3. Cosine similarity.
4. Hamming distance.

Euclidean distance: The straight-line distance b/w two points in Euclidean space. $d = \sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$

Manhattan distance: Manhattan distance (also called L1 distance or taxicab distance) is the distance b/w two points measured along axes at right angles.

$d = |(x_2-x_1)+(y_2-y_1)|$.

Cosine similarity: Measures the cosine of the angle b/w two non-zero vectors, indicating how similar they are regardless of their magnitude.

$$\text{Cosine similarity} = \frac{A \cdot B}{\|A\|\|B\|}$$

Hamming distance: It is the distance b/w two strings of equal length is the no. of positions at which corresponding symbols are different.

* It is mainly used in binary strings, DNA sequences & error detection/correction in coding theory.

Problems:

1. Find the Euclidean & Manhattan distance between two points.
$P_1(3, 2)$ & $P_2(4, 1)$.

Sol: ED $= \sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$

$= \sqrt{(4-3)^2+(1-2)^2} = \sqrt{1+1} = \sqrt{2} = 1.414$.

MD $= |(x_2-x_1)+(y_2-y_1)| = |(4-3)+(1-2)|$

$= |(1)+(-1)| = |0|$.

2. A triangle has 3 vertices A(2,3), B(5,7), C(8,1). Find the length of longest side of triangle. compare the value of both distances.

Sol: ED of AB. A$(2,3)$ B$(5,7)$

ED $= \sqrt{(5-2)^2+(7-3)^2} = \sqrt{(3)^2+(4)^2} = \sqrt{9+16}$

$= \sqrt{25} = 5$.

ED of BC. B$(5,7)$ C$(8,1)$

ED $= \sqrt{(8-5)^2+(1-7)^2} = \sqrt{(3)^2+(-6)^2} = \sqrt{9+36}$

$= \sqrt{45} = 6.708$

ED of ~~CA~~ CA. C$(8,1)$ A$(2,3)$

ED $= \sqrt{(2-8)^2+(3-1)^2} = \sqrt{(6)^2+(2)^2} = \sqrt{36+4} = \sqrt{40}$

$= 6.324$.

Longest length of ED is BC $= 6.708$.

MD of AB. A(2,3), B(5,7)

MD = |(5-2)+(7-3)| = |3+4| = |7|

MD of BC. B(5,7) C(8,1)

MD = |(8-5)+(1-7)| = |3+(-6)| = |3-6|
= |-3| = 3.

MD of CA. C(8,1) A(2,3)

MD = |(2-8)+(3-1)| = |(-6)+2| = |-4| = |4|.
= 4.

longest side = 7 (AB).

Data Mining & functionalities:

It is a tools as a technical methodology to detect information from huge datasets. It has 2 categories:

* Descriptive data mining.
* Predictive data mining.

Descriptive data Mining:

* Used for finding patterns & relationship in data.

Cluster analysis - Group of data points.

Association Rule mining - Identify relationship b/w variables.

Visualization - To represent visual.

Predictive data Mining: It is concerned with developing models which can predict the future behaviour.

Common techniques:

1. Decision tree: Used to create a model that can predict the value of target variable. It is used for classification.
2. Neural Networks: to create a model to recognize

patterns. Eg: Speech Recognition.

3. Regression: Analysis: To create model & predicts the value of target variable.

Data preparation algorithms: It is the process of making raw data ready for after processing and analysis. The key methods are to collect, clean & label raw data in a format suitable for machine learning algorithms, followed by data exploration & visualization.

→ Data preparation Process:

step 1: Data collection:

Gather data from various sources such as:

* Databases.
* Sensors and IoT devices.
* Websites & APIs.
* Surveys, and forms.
* Transaction records.

step 2: Data Integration: Combine data from multiple sources into a unified dataset.

Remove duplicate records.

* Resolve inconsistencies.
* Standardize formats.

step 3: Data cleaning:

Improve data quality by handling.

* Missing values (mean, median, mode, imputation)
* Duplicate records.
* Outliers.

step 4: Da
Convert c
learnin
* Normal
* Standa
* Aggreg
* Encodi

step 5: Feat
from exist
Eg: Age
total pu

step-6: fea
features a
benefits:

step 7: Ha
segmentan
* Oversam
* Under
* class

step 8: D
* Trainin
* Valida
* Test s

step 9: Data
consis
Techniqu
* dimens
* Sampli
* Featu

Step 4: Data transformation:
Convert data into a format suitable for machine learning.
* Normalisation.
* Standardization.
* Aggregation.
* Encoding categorical variables. Eg: gender: Male/female → 0/1.
* Scaling numerical values to a common range.

Step 5: Feature Engineering: Create new useful features from existing data.
Eg: Age → Age Group. Date → Day, month, year.
total purchase value = Quantity x Price.

Step-6: feature selection: Select the most relavant features and remove irrelavant or redundant ones.
benefits: * Reduces model complexity.
* Improves accuracy.
* Decreases training time.

Step 7: Handling Imbalanced data: When one class has significantly more samples than another.
* Oversampling. (e.g. SMOTE).
* Undersampling.
* class weighting.

Step 8: Data splitting: Divide data into seperate sets:
* Training Set (typically 70-80%).
* Validation set (10-15%)
* Test set (10-15%).

Step 9: Data Reduction (Optional): Reduce dataset size while preserving information.
Techniques:
* dimensionality reduction.
* Sampling.
* Feature extraction.

Step 10: Final Verification: Before training;

* Check data quality.
* Verify feature distributions.
* Ensure no data leakage.
* Confirm correct train-test separation.

Min-Max Normalization: It is the most common way of normalizing the value. If minimum value of the feature is 0 model gets transformed to 0 value, the maximum value gets transformed to 1. So it lies in the range between 0 to 1.

$$x_{new} = \frac{x - x_{min}}{x_{max} - x_{min}}$$

Z-Score Normalization (or) Inter Quartile method (IQM):

It is the strategy of normalizing the data that avoids ~~outlier~~ outlayer (outlier) issue.

Steps to find Z-score:

1. Calculate mean (μ) & standard deviation (σ).
2. For each data point compute z-score.

$$z = \frac{x - \mu}{\sigma}$$

$x \rightarrow$ original raw data point
$\mu \rightarrow$ Mean,
$\sigma \rightarrow$ S.D.

3. If $z > +3$ (or) $z < -3$, otherwise point is a outlier.

Qums:

1. Dataset = {1000, 1200, 1100, 1300, 1250, 1280, 1050, 1400, 5000} using z score method Find whether 5000 Rs an outlier.

Sol: I) Mean μ = 1620.

II) S.D: difference from mean:

= {1620-1000, 1620-1200, 1620-1100, 1620-1300, 1620-1250, 1620-1280, 1620-1050, 1620-1400, 1620-5000}

= {620, 420, 520, 320, 370, 340, 570, 220, 3380}

Square of difference:

{384400 + 176400 + 270400 + 102400 + 136900 + 115600 + 324900 + 48400 + 11424400}.

Mean of square of difference:

$= \frac{12740440}{9} = 1,442,644.44$

Variance = $\sigma^2 = 1,442,644.44$

S.D = $\sqrt{1,442,644.44} = 1201.10$

2) Z-score = $\frac{x-\mu}{\sigma} = \frac{5000-1620}{1201.10} = \frac{3380}{1201.10}$

$= 2.81$

∴ 5000 Rs not an outlier.

2. The attributes of age in increasing order as 13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 25, 30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70.

Find: a) Frequency table

b) Mean

c) SD. try to identify the outliers if any.

Sol: a) Frequency table:

Numbers	frequency.
13	1
15	1
16	2
19	1
20	2
21	1
22	2
25	4
30	1
33	2
35	4
36	1
40	1
45	1
46	1
52	1
70	1

Mode = 25, 35.

b) Mean = 29.96

c) S.D: Difference from mean:

= {16.96, 14.96, 13.96, 13.96, 10.96, 9.96, 9.96, 8.96, 7.96, 7.96, 4.96, 4.96, 4.96, 4.96, -0.04, -3.04, -3.04, -5.04, -5.04, -5.04, -5.04, -6.04, -10.04, -15.04, -16.04, -22.04, -40.04}.

Square of difference:

= {287.64, 223.80, 194.88, 194.88, 120.12, 99.20, 99.20, 80.28, 63.36, 63.36, 24.60, 24.60, 24.60, 24.60, 0.0016, 9.24, 9.24, 25.40, 25.40, 25.40, 25.40, 36.48, 100.80, 226.20,

257.28, 485.76, 1603.204.

Mean of square of difference:

$= 1300 + 243.60 + 2811.32 = \frac{4354.92}{27}$

$= 161.29$

Variance $= \sigma^2 = \sqrt{161.29}$

S.D $= \sqrt{161.29} = 12.70$.

Z-score: $Z = \frac{x - \mu}{\sigma} = \frac{13 - 29.96}{12.70} = -1.33$

$z = \frac{15 - 29.96}{12.70} = -1.17$

$z = \frac{16 - 29.96}{12.70} = -26.66$

$z = \frac{19 - 29.96}{12.70} = -0.86$

$z = \frac{20 - 29.96}{12.70} = -0.78$

$z = \frac{21 - 29.96}{12.70} = -0.70$

$z = \frac{22 - 29.96}{12.70} = -0.62$

$z = \frac{25 - 29.96}{12.70} = -0.39$

$z = \frac{30 - 29.96}{12.70} = 0.003$

$z = \frac{33 - 29.96}{12.70} = 0.23$

$z = \frac{35 - 29.96}{12.70} = 0.39$

$z = \frac{36 - 29.96}{12.70} = 0.47$

$z = \frac{40 - 29.96}{12.70} = 0.79$

$z = \frac{45 - 29.96}{12.70} = 1.18$

$z = \frac{46 - 29.96}{12.70} = 1.26$

$z = \frac{52 - 29.96}{12.70} = 1.73$

$z = \frac{70 - 29.96}{12.70} = 3.15$.

70 is the outlier.

3. Dataset = {200, 300, 400, 600, 1000} min-max normalization by setting min = 0 and max = 1. Use z-score normalization, decimal scaling standard deviation,

Sol: Min-max normalization range (0, 1)

$$x_{new} = \frac{x - x_{min}}{x_{max} - x_{min}} \qquad x_{min} = 200 \quad x_{max} = 1000$$

$$x_{200} = \frac{200-200}{1000-200} = \frac{0}{800}, \quad x_{300} = \frac{300-200}{1000-200} = \frac{100}{800} = \frac{1}{8}$$

$$x_{400} = \frac{400-200}{1000-200} = \frac{200}{800} = \frac{1}{4}, \quad x_{600} = \frac{600-200}{1000-200} = \frac{400}{800} = \frac{1}{2}$$

$$x_{1000} = \frac{1000-200}{1000-200} = \frac{800}{800} = 1.$$

Dataset	200	300	400	600	1000
Min-Max value	0	0.125	0.25	0.5	1

Decimal scaling of 200 = $\frac{200}{1000} = 0.2$

" " 300 = $\frac{300}{1000} = 0.3$

" " 400 = $\frac{400}{1000} = 0.4$

" " 600 = $\frac{600}{1000} = 0.6$

" " 1000 = $\frac{1000}{1000} = 1.$

Z-Score: $Z = \frac{x - \mu}{\sigma}$.

Mean = 500. $\Rightarrow \mu$.

Difference from mean:

$\{-300, -200, -100, 100, 500\}$.

Square of difference: $\{90000, 40000, 10000, 10000, 250,000\}$.

Mean of square of differences: 80,000.

Variance $= \sigma^2 = 80000$.

standard deviation $= \sqrt{80000} = 282.84$

z-score: $z = \frac{200-500}{282.84} = \frac{-300}{282.84} = -1.06$.

$z = \frac{300-500}{282.84} = \frac{-200}{282.84} = -0.70$

$z = \frac{400-500}{282.84} = \frac{-100}{282.84} = 0.35$

$z = \frac{600-500}{282.84} = 0.35$, $z = \frac{1000-500}{282.84} = 1.76$